

Julia V.E. Kim

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Education

Massachusetts Institute of Technology

PH.D. OPERATIONS RESEARCH

Boston, US

Sept. 2025 - Jun. 2030 (expected)

- **Cumulative GPA:** 5.0/5.0
- **Research Interests:** AI safety, AI Evaluation, Multimodal AI, Optimisation

University of Cambridge, Trinity College

GRADUATE STUDIES IN MATHEMATICS

Cambridge, UK

Oct. 2024 - Jun. 2025

- **Scholarship:** Trinity Studentship in Mathematics
- **Thesis:** Applying Contrastive Learning to Stellar Spectra

University of Toronto

BSc (HONOURS), PHYSICS, MATHEMATICS & STATISTICS

Toronto, CA

Sep. 2019-Jun.2024

- **Cumulative GPA: 4.00/4.00, Average: 97.6%, Rank: 1/16,000**
- **Scholarship:** Rose Sheinin Award for *top female student* in science at UofT

Professional Experience

Data Science Intern

THE TRADE DESK: AI LAB

London, UK

Jun. 2025 - Aug. 2025

- Built end-to-end GPID validation and scoring pipeline in **PySpark** on **AWS S3**, processing 5.3M+ bidstream records for ad targeting
- Designed signal validation and data quality workflows for ad targeting, supporting marketplace quality research

Astrophysics Machine Learning Researcher

UNIVERSITY OF TORONTO

Toronto, CA

May 2024-present

- Designed and trained deep convolutional neural networks in **PyTorch** to predict stellar parameters with high precision and accuracy
- Developed self-supervised learning models, successfully embedding cross-modal data into shared latent space
- Achieved an R^2 accuracy of over 95% on one-shot stellar property estimation tasks, outperforming traditional baselines by 19%

Biophysics Machine Learning Researcher

UNIVERSITY OF TORONTO

Toronto, CA

May 2023-May 2024

- Built a stochastic model in **TensorFlow** to simulate receptor signalling pathways with cross-talk and pleiotropy
- Applied information theory and statistical inference to investigate the impact of feedback and molecular noise on signalling specificity
- Improved model accuracy by 30% through hyperparameter tuning, feature selection and optimising neural network architecture

Dark Matter Particle Physics Researcher

SUPERCDCMS SNOLAB & UNIVERSITY OF TORONTO

Toronto, CA

Sep. 2021-Apr.2022

- Developed likelihood-based statistical models in **C++** to establish sensitivity projections for detecting dark matter signals
- Performed over 1,000 Monte Carlo simulations to generate test statistics for LIP detection at 95% confidence level
- Leveraged **Roofit** and **RooStats** to estimate an upper limit of 120 signal events, validating results with χ^2 and Gaussian distributions

Skills

Languages: Python, R, MATLAB, Julia, Bash, LaTeX

Spoken: English (native), French (native), Turkish (intermediate, studying at Harvard)

Libraries/Frameworks: PyTorch, TensorFlow, Numpy, Pandas, Matplotlib, Scikit-learn, Seaborn

Tools: PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, Hugging Face, Git

Publications

1. Julia V.E. Kim & John R. Percy, "Pulsating Red Giants in a Globular Cluster: ω Centauri", 2022, JAAVSO, 50, 2. [Link](#) 🌐.
2. Natasha Leclerc, Julia V.E. Kim et al. 'Ba Depletion in Coastal Waters Points to Planktonic Phytoplankton Intensification'. (in progress)
3. Julia V.E. Kim. 'Patterns and Trends in Endangered Species Listing Heavily Biased Towards Imperiled Vertebrates with High Utility Value to Humans.' Featured in Alexander, R. (2024) *Telling Stories with Data with Applications in R*. Milton: CRC Press LLC. [Link](#). 🌐
4. Julia V.E. Kim. 'Significant Declines in Standardised Test Scores due to COVID-19 School Closures Disproportionately Affect Vulnerable Students.' Featured in Alexander, R. (2024) *Telling Stories with Data with Applications in R*. Milton: CRC Press LLC. [Link](#). 🌐

Projects

- Using a Deep Deterministic Gradient Policy (DDPG) to Control a Two-Joint Arm

MACHINE LEARNING COURSE PROJECT

University of Toronto

Mar.-Apr. 2023

Implemented DDPG reinforcement learning algorithm in **MATLAB** to optimise the motion of a two-joint robotic arm

Developed and trained neural networks with ReLU activation, stochastic gradient descent and Adam optimisation

Achieved improved policy returns over 1, 000 rollouts, by leveraging test policies and minibatch learning with replay buffer
- Learning MNIST Handwritten Digits with a Retinotopic Map

MACHINE LEARNING COURSE PROJECT

University of Toronto

Feb.-Mar. 2023

Implemented neural network to recognise digits, using backpropagation with Adam and Nesterov in MATLAB

Developed 3-layer mapnet with ReLU activations, applying softmax for classification and training through minibatch learning

Reduced classification error from 480 to below 170 incorrect guesses over 10 epochs across 60, 000 training and 10, 000 test examples

Awards & Honours

- University of Toronto Dean’s List Scholar, University of Toronto

2020-2024
- Rose Sheinin Award, University of Toronto

Jun. 2024
- St Micheal’s College Achievement Award in Maths and Physical Sciences, University of Toronto

Jun. 2024
- Governor General Silver Medal Nominee, University of Toronto

Jun. 2024
- University of Toronto Excellence Award, University of Toronto

Jun. 2023, Jun. 2024
- Trinity Studentship in Mathematics, University of Cambridge

Apr. 2024
- John J Sebisty Scholarship, University of Toronto

Mar. 2024
- David and Louise Fraser Scholarship, University of Toronto

Mar. 2024
- Hymie and Roslyn Mida Student Award in Theoretical Physics, University of Toronto

Mar. 2024
- Lucia Colavita Memorial Scholarship, University of Toronto

Jun. 2023
- Jackman Scholars-in-Residence Award, Victoria College

Jun. 2023
- The 3T0 M.&P. and Associates Scholarship, University of Toronto

Dec. 2021, Feb. 2023
- Patricia Massel 5T1 Bursary, University of Toronto

Jun. 2022
- NSERC USRA and University of Toronto Chair Scholar, NSERC

May 2022
- St Micheal’s College In Course Scholarship, University of Toronto

Mar. 2022
- University of Toronto In-Course Scholarship, University of Toronto

Sep. 2021
- University of Toronto Scholar’s Award, University of Toronto

Aug. 2021
- Samuel Beatty In-Course Scholarship, University of Toronto

Jan. 2021
- Joseph Alfred Whealy In-course Scholarship

Teaching

- University of Cambridge, Examinations Invigilator

Oct. 2024-present
- University of Toronto, MAT137Y1: Calculus! TA

Sep. 2020-May 2021
- University of Toronto, MAT233: Linear Algebra I workbook editor

May-Jul. 2020

Talks

1. ‘Applying Self-Supervised Learning to the Analysis of Stellar Spectra’, SURP Presentation Day. University of Toronto (Aug. 2024).
2. ‘Ba-Ca and stable C isotope ratios in *C. compactum* suggest increase in Artic primary productivity driven by sea loss and higher nutrient input from permafrost melt and terrestrial run-off’, ASLO Conference, Madison, Wisconsin, USA (Jun. 2024).
3. ‘Replication Analysis of COVID-19 Test Scores’, DSI Reproducibility Challenge. Data Science Institute, University of Toronto (May 2024).
4. ‘Investigating *C. compactum* as an Artic Erosion Proxy’, UTM Indicium Talk. University of Toronto (Jul. 2023).
5. ‘Generalising $M_{T_2}^{DC}$ for Decay Chains and Mass Measurements’, ATLAS Talk. University of Toronto (Jul. 2022).

Outreach

- MIT OGE, Graduate Community Fellow

August 2026-present
- MIT Student INFORMS Chapter, Vice President

Sep. 2025-present
- Trinity College Cambridge, APPD Programme Assistant

Oct. 2024-May 2025
- Girls in STEAM, Vice President of Content

Jan. 2023-May 2024
- University of Toronto Machine Intelligence Team, Academic Coordinator

Jan. 2023-May 2024
- University of Toronto Aerospace Team, Structures Division Assistant

Jun. 2021-May 2023
- The Varsity, University of Toronto Newspaper Science Writer

Jun. 2022- Jun.2023

Media

Kaster, J. (2024) *St. Mike’s Julia Kim wins top U of T science award*, St. Micheal’s College, University of Toronto. [Link](#)